

Aritra Bhattacharya

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Education

Texas A&M University

PhD in Entomology

August 2023 – Present

College Station, Texas, USA

Maulana Abul Kalam Azad University of Technology, WB

MSc in Biotechnology

November 2020 – June 2022

Haringhata, West Bengal, India

Ramakrishna Mission Vivekananda Centenary College

BSc in Zoology

July 2016 – October 2020

Khordaha, West Bengal, India

Research Experience

Evolution of resistance under divergent rate of pathogen selection

Ashoka University, Sonapat, Haryana, India (February 2022 – June 2022)

- This was my MSc dissertation thesis. I did some experiments to check the rate of evolution of resistance against different pathogens in *Tribolium* beetles.

COPD: Making Sense of Existing GWAS Findings in Indian Context

MAKAUT, WB, Haringhata, India (December 2021 - February 2022)

- It was a small research project during my MSc, We jotted down the genes and SNPs responsible for COPD in the Indian population from 1456 variants & did pathway analysis to check their functions that might be relevant for COPD development.

Temporal diversity of urban butterflies: A case study

RKMVC College, Khordaha, WB, India (November 2016 - January 2020)

- I did this project during my BSc. We sampled butterflies in and around our college campus. We did a checklist of butterflies and also observed how their community is changing across seasons with vegetation pattern.

Professional Experience

PhD student

Motta Lab, Department of Entomology, Texas A&M University

Summer 2025 – present

College Station, TX, USA

- Being a PhD student at the lab, my roles are including but not limited to conducting experiments, analyzing and visualizing my data and presenting them in lab meetings and conferences, collaborating with current lab members in their projects, and mentoring undergraduates.

Graduate Research Assistant

Magalhaes Lab, Department of Entomology, Texas A&M University

Summer 2024 – Spring 2025

College Station, TX, USA

- Being a PhD student at the lab, my responsibilities included doing the safety checks every weeks, maintaining the mosquito populations, keeping track of the stock of reagents and other materials apart from doing experiments for my project.

Graduate Teaching Assistant

Summer 2025 – Present

Dept. of Entomology, Texas A& M University, College Station, TX, USA

- I have been teaching the lab sections of FIVS 215 (Forensic microscopy) in the Department of Entomology this Fall (Fall 2026), here I teach basics of polarized light microscopy and their applications in forensics by showing presentations and demonstrations and helping the students in their lab exercises. I have taught the lab sections of FIVS 210 (Forensic photography) in Fall 2025 and Spring 2026, where I taught the basics of photography techniques using a DSLR camera and their applications in forensics.

Fall 2023 – Spring 2024

Department of Biology, Texas A& M University, College Station, TX, USA

- I have taught the lab courses of BIOL 111 (Introductory Biology I) in Fall 2023 & BIOL 319 (Anatomy and Physiology I) in Spring 2024. This role requires me to give a background of the topic of the lab to the undergraduate students, supervise them while they are doing their lab exercises, grade their laboratory assignments, set up questions and proctor Lab exams and grading them. I have taught the lab sections of FIVS 210 (Forensic Photography) in the Department of Entomology in Fall 2025 as well.

Project Assistant

July 2022 – March 2023

Evolutionary Immunology Lab, Ashoka University

Sonepat, Haryana, India

- I studied sperm competition abilities of *Tribolium* beetles and the role of chronic malnutrition in disease transmission in *Drosophila* as a research assistant after completing my MSc. This position required me to do experiments, record, analyze and present data & maintain populations of the model organisms.

Skills

Computer Languages: R (intermediate), Python (beginner), Bash (beginner), HTML(intermediate).

Wet Lab techniques: DNA extraction, RNA extraction, PCR, gel electrophoresis, spectrophotometry, high resolution camera operated microscopy, basic microbial techniques, field sampling techniques.

Softwares: Basic computer applications like Microsoft Office, SPSS, PAST, different bioinformatic tools.

Other skills: Digital photography and photo editing, digital art, music production.

Presentations

- Student Research Week, 'Disentangling the Mechanistic Principles of Microbiota-Mediated Detoxification in Honey Bees', (03/17/2026, College Station, TX, USA) **Oral**
- 4th Annual Conference of the Indian Society for Evolutionary Biologists, 'Revolutionising Darwinian Synthesis: Fresh Perspectives and Future Challenges.', (02/09/2023-02/11/2023, Ahmedabad, India) **Poster**
- Manufacturing (Level 1) & COVID-Food Safety Supervisor, (02/20/2021-02/21/2021, Kolkata, India) **Poster**[Won technical brilliance award]

- 6th Asian Lepidoptera Conservation Symposium, (09/17/2019-09/21/2019, Kolkata, India) **Poster**
- ALLERCON-2017, The 7th National Conference on Respiratory Allergy and Immunology, (01/14/2018, Kolkata, India) **Poster** [Won best presenter award]
- National Conference on Status of Invasive Alien Species in India, (12/14/2017-12/15/2017, Kolkata, India) **Poster**
- National Seminar on Forest Resources: Conservation Issues and Management, (09/09/2017, Kolkata, India) **Poster**

Certifications

AI0101EN: AI for Everyone: Master the Basics	4 weeks
Certificate ID: 6fd8789d1ae146348a4bda736f3696d6	Platform: edX
PY0101EN: Python Basics for Data Science	5 weeks
Certificate ID: 5cc9d19334e94a9682fa0bc969f726c0	Platform: edX
PH125.2x: Data Science: Visualization	8 weeks
Certificate ID: a40f449f0e534960b7ef8fe3f572ed56	Platform: edX
LFS101x: Introduction to linux	14 weeks
Certificate ID: 4a6c484ebde6412cb68fba71e4ba2828	Platform: edX

Publications

- Dutta, T., **Bhattacharya, A.** 2022. Chronic Obstructive Pulmonary Disease (COPD): Making Sense of Existing GWAS Findings in Indian Context. *Int J Biomed Clin Anal.*, **2(1)**:29-37
- **Bhattacharya, A.**, Chakraborty, S., Das, K.L., Saha, A., Sardar, S., Mandal, B., Mukherjee, A. and Mitra, B. 2018. Temporal diversity of urban butterflies: A case study in the campus of Ramakrishna mission Vivekananda centenary College, Rahara, West Bengal. *The Pharma Innovation Journal*, **7(12)**: 395-400.

Publications in abstracts

- **Bhattacharya, A.**, Motta, E. 2026/16/03 to 2026/20/03. Disentangling the Mechanistic Principles of Microbiota-Mediated Detoxification in Honey Bees. *Student Research Week, 2026'* Pp- 60
- **Bhattacharya, A.**, Bose, J., Khan, I. 2023/09/02 to 2023/11/02. Linking evolution under chronic malnutrition, costs of infection and disease transmission. *4th Annual Conference, Indian Society of Evolutionary Biologists, 'Revolutionising Darwinian Synthesis: Fresh Perspectives and Future Challenges.'* Pp- 87
- **Bhattacharya, A.**, Das, S., Ghosh, D., Sardar, S., Mandal, B., Mukherjee, A., Mitra, B. 2019/17/09 to 2019/21/09. Community foraging of butterflies on *Lantana camara* – partition or overlapping. *6th Asian Lepidoptera Conservation symposium.* Pp- 90

- **Bhattacharya, A.**, Chakraborty, S., Das, K.L., Sardar,S., Mandal, B. and Murkherjee, A. 2018/14/01. Alternative proposal for combating the pollen-induced allergy by introducing non allergic butterfly parks in urban area- a case study. *ALLERCON-2017, The 7th National Conference on Respiratory Allergy and Immunology*. Pp-36
- Chakraborty, S., Das, K.L., **Bhattacharya, A.**, Saha, A., Sardar,S., Mandal, B. and Murkherjee, A. 2017/14/12 & 2017/15/12. Temporal shifting in communities of butterflies in a suburban area and it's relation with social plantation. *National Conference on Status of Invasive Alien Species in India*. Pp-59.
- **Bhattacharya, A.**, Saha, A., Chakraborty, S., Das, K.L., Sardar,S., Mandal, B. and Murkherjee, A. 2016/09/09. Small scale social foresting in suburban area for upholding the temporal variations of butterflies. *Nat. Sem. On Forest Resources: Conser. Issues and Mgmt*. Pp- 30.

Grants

EKTA Incubation Centre-TBI (MAKAUT,WB): 45000INR for ecological restoration.
(Declined)